Experiment No.4 Advanced Data Visualization Matplotlib and seaborn

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/archive.zip")

print(df.head())
print(df.info())
print(df.describe())
```

```
3       4  US-2016-108966  11/10/2016  18/10/2016  Standard Class    SO-20335
4       5  US-2016-108966  11/10/2016  18/10/2016  Standard Class    SO-20335

     Customer Name    Segment        Country             City       State  \
0      Claire Gute   Consumer  United States        Henderson    Kentucky
1      Claire Gute   Consumer  United States        Henderson    Kentucky
2  Darrin Van Huff  Corporate  United States      Los Angeles  California
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale     Florida
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale     Florida

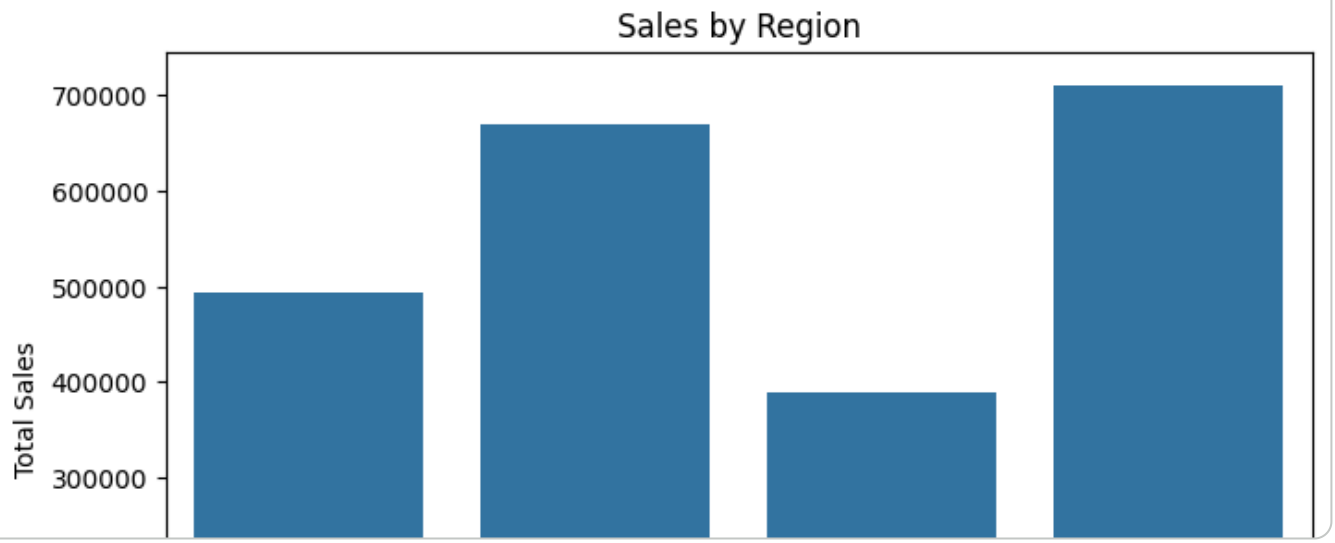
   Postal Code Region       Product ID         Category Sub-Category  \
0      42420.0  South  FUR-BO-10001798        Furniture    Bookcases
1      42420.0  South  FUR-CH-10000454        Furniture       Chairs
2      90036.0   West  OFF-LA-10000240  Office Supplies       Labels
3      33311.0  South  FUR-TA-10000577        Furniture       Tables
4      33311.0  South  OFF-ST-10000760  Office Supplies      Storage

                                        Product Name     Sales
0                  Bush Somerset Collection Bookcase  261.9600
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
4                    Eldon Fold 'N Roll Cart System   22.3680
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9800 non-null   int64
 1   Order ID       9800 non-null   object
 2   Order Date     9800 non-null   object
 3   Ship Date      9800 non-null   object
 4   Ship Mode      9800 non-null   object
 5   Customer ID    9800 non-null   object
 6   Customer Name  9800 non-null   object
 7   Segment        9800 non-null   object
 8   Country        9800 non-null   object
 9   City           9800 non-null   object
 10  State          9800 non-null   object
 11  Postal Code    9789 non-null   float64
 12  Region         9800 non-null   object
 13  Product ID     9800 non-null   object
 14  Category       9800 non-null   object
 15  Sub-Category   9800 non-null   object
```

```
             Row ID   Postal Code         Sales
count  9800.000000   9789.000000   9800.000000
mean   4900.500000  55273.322403    230.769059
std    2829.160653  32041.223413    626.651875
min       1.000000   1040.000000      0.444000
25%    2450.750000  23223.000000     17.248000
50%    4900.500000  58103.000000     54.490000
75%    7350.250000  90008.000000    210.605000
max    9800.000000  99301.000000  22638.480000
```

```
region_sales = df.groupby("Region")["Sales"].sum().reset_index()

plt.figure(figsize=(8,5))
sns.barplot(data=region_sales, x="Region", y="Sales")
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.show()
```

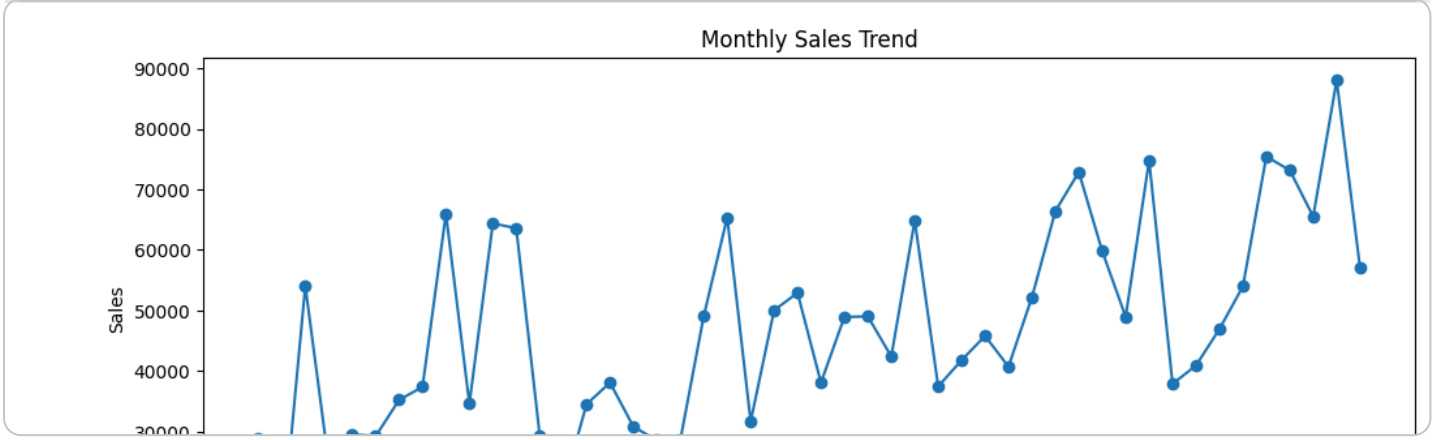



```
df["Order Date"] = pd.to_datetime(df["Order Date"], format="mixed", errors="coerce")

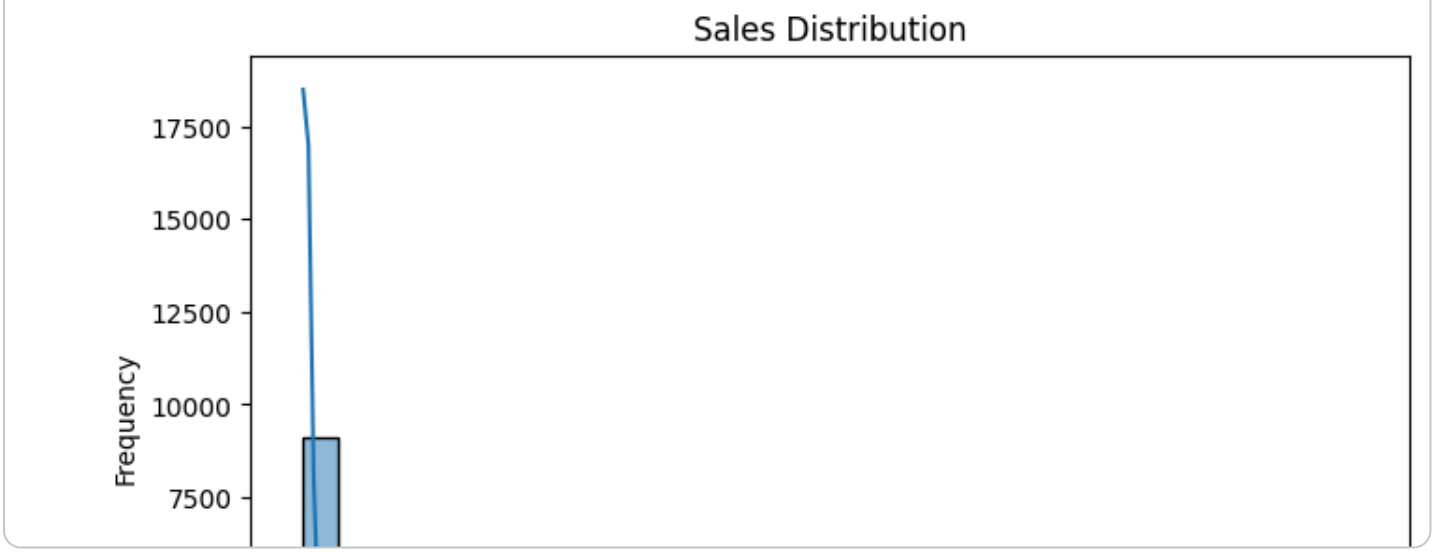
df = df.dropna(subset=["Order Date"])

monthly_sales = df.groupby(df["Order Date"].dt.to_period("M"))["Sales"].sum()

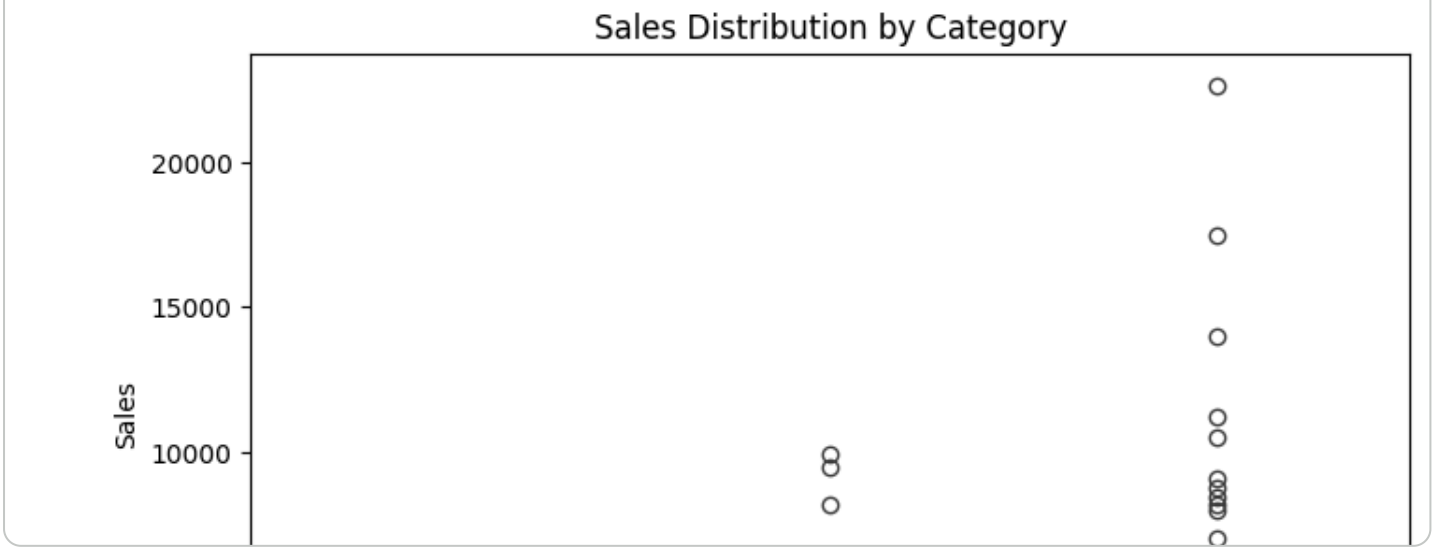
plt.figure(figsize=(12,5))
plt.plot(monthly_sales.index.astype(str), monthly_sales.values, marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.histplot(df["Sales"], bins=30, kde=True)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

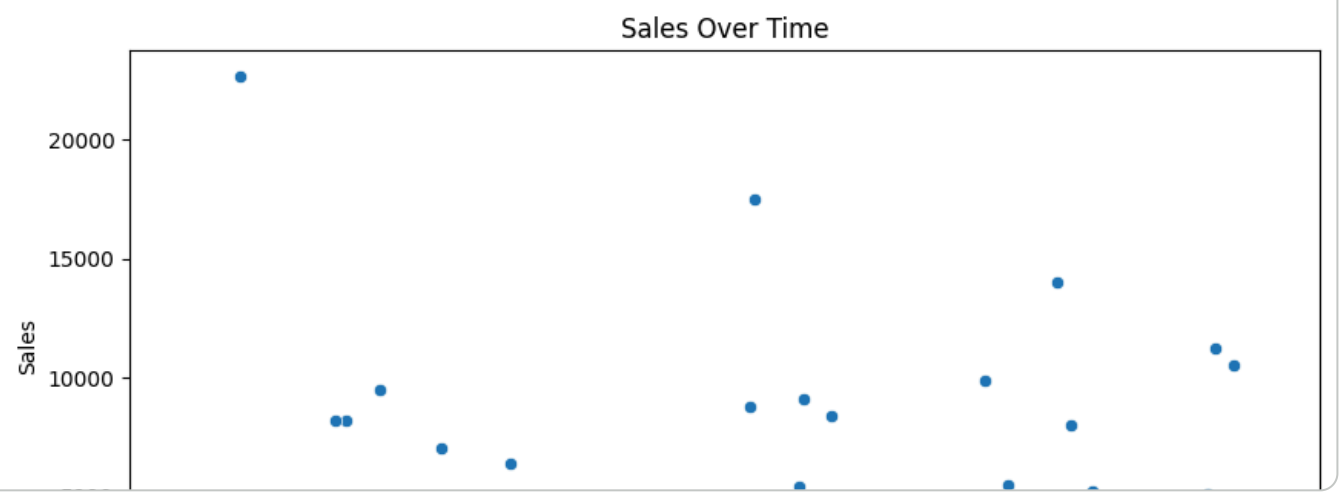



```
plt.figure(figsize=(8,5))
sns.boxplot(data=df, x="Category", y="Sales")
plt.title("Sales Distribution by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```



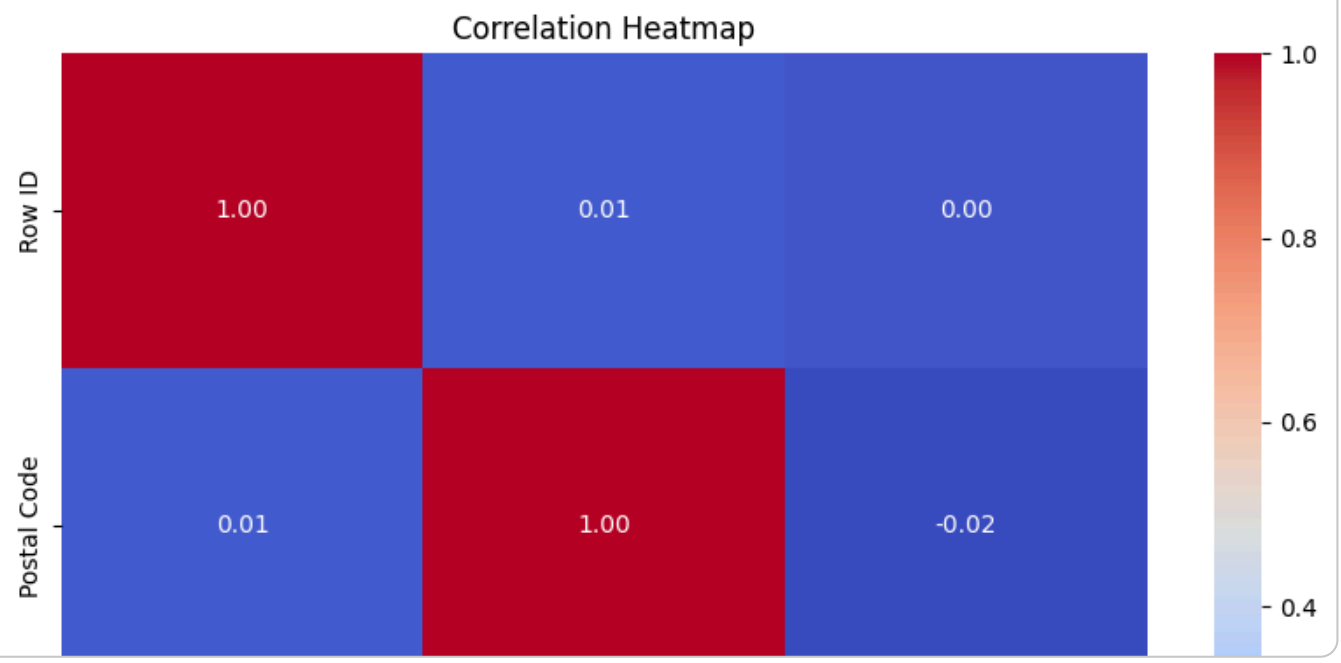
```
df["Order Date"] = pd.to_datetime(df["Order Date"], format="mixed", errors="coerce")

plt.figure(figsize=(10,5))
sns.scatterplot(data=df, x="Order Date", y="Sales")
plt.title("Sales Over Time")
plt.xlabel("Order Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```



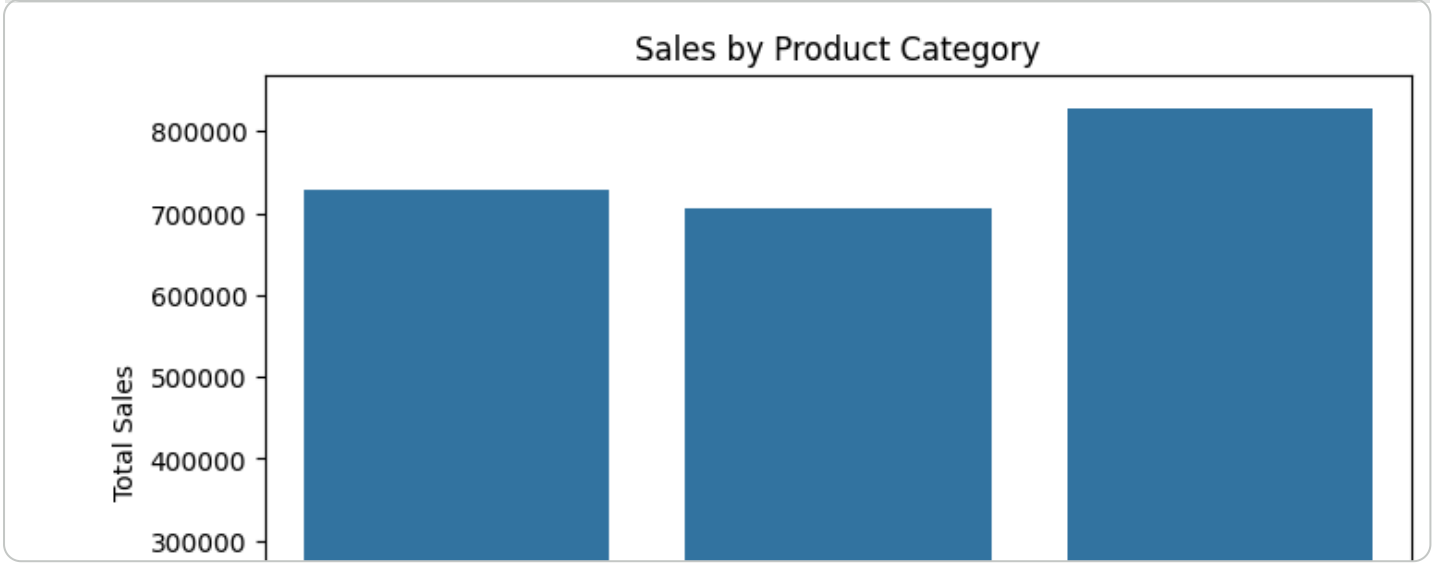

```
numeric_df = df.select_dtypes(include="number")
correlation = numeric_df.corr()

plt.figure(figsize=(10,7))
sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



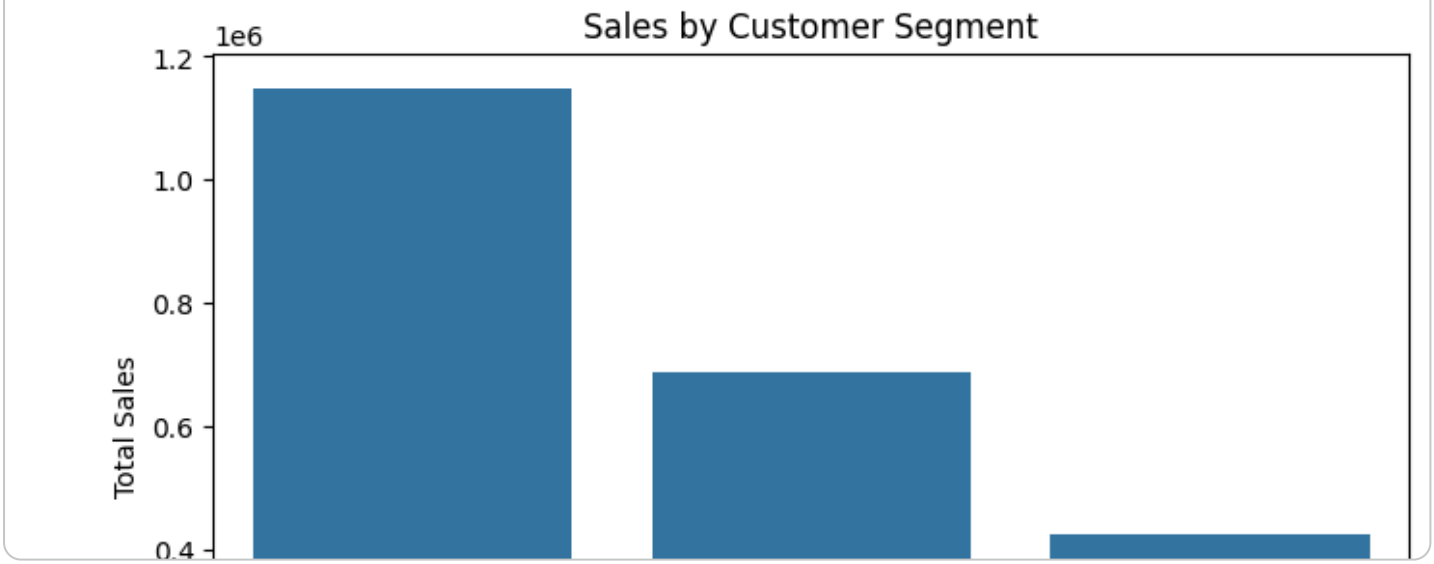

```
category_sales = df.groupby("Category")["Sales"].sum().reset_index()

plt.figure(figsize=(8,5))
sns.barplot(data=category_sales, x="Category", y="Sales")
plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```



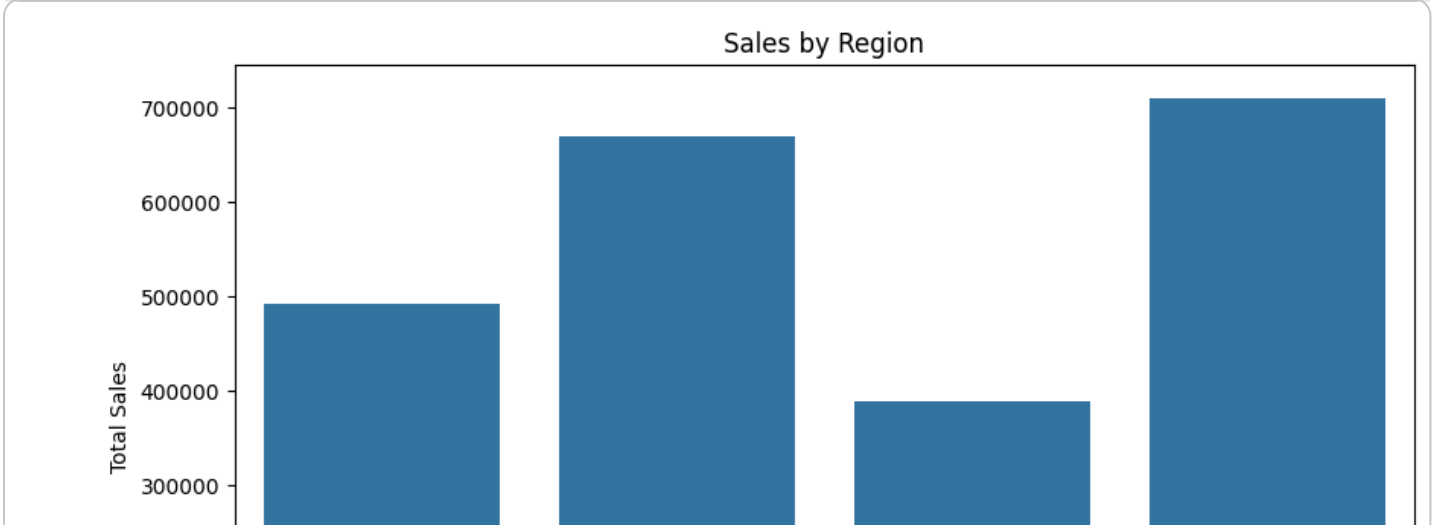

```
segment_sales = df.groupby("Segment")["Sales"].sum().reset_index()

plt.figure(figsize=(8,5))
sns.barplot(data=segment_sales, x="Segment", y="Sales")
plt.title("Sales by Customer Segment")
plt.xlabel("Segment")
plt.ylabel("Total Sales")
plt.show()
```




```
region_data = df.groupby("Region")["Sales"].sum()

plt.figure(figsize=(10,6))
sns.barplot(data=region_data, x="Region", y="Sale
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.show()
```


Sales by Region
Total Sales
700000
600000
500000
400000
300000